Fastsae: High-Performance Frequentist and Bayesian Small Area Estimation in R

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Abstract Small area estimation (SAE) plays a pivotal operational role in official statistics, enabling the production of reliable disaggregated socioeconomic, health, and demographic indicators when domain-specific sample sizes are small or zero. However, practitioners in national statistical offices (NSOs) face a triple computational and operational barrier: frequentist spatial and spatio-temporal models with bootstrap mean squared error (MSE) estimation suffer from severe memory bottlenecks and slow runtimes in interpreted languages; non-Gaussian indicators (such as rates, skewed expenditures, and counts) traditionally require computationally demanding Markov chain Monte Carlo (MCMC) algorithms that impede routine production; and model estimates often fail to aggregate to published higher-level totals, violating institutional consistency. We introduce fastsae, an R package engineered for high-performance frequentist and Bayesian small area estimation. On the frequentist side, fastsae implements optimized C++17 routines via RcppArmadillo paired with OpenMP multi-threading, accelerating the classical Fay-Herriot, spatial Fay-Herriot, spatio-temporal Fay-Herriot, and unit-level Battese-Harter-Fuller models alongside their analytical and bootstrap MSE estimators. On the Bayesian side, fastsae provides a unified hierarchical Bayes (HB) interface across six distribution families (Gaussian, Beta, Gamma, Binomial, Poisson, Negative Binomial), fitted by Integrated Nested Laplace Approximations (INLA) by default or by MCMC in Python (JAX/NumPyro) through interchangeable engine objects. The likelihood dispersion of each area is fixed at its design-based sampling variance, the remaining hyperparameters have documented default priors that were calibrated by simulation, and the models accommodate modern spatial priors (BYM, scaled BYM2, Leroux CAR, and SLM) as well as Knorr-Held space-time interaction dynamics. To bridge estimation and dissemination, the package features a two-stage hierarchical benchmarking engine for simultaneous provincial and national calibration, an automated reliability classification engine aligned with NSO standards, multi-model concordances, automated ggplot2 spatial mapping, and multi-sheet spreadsheet exports. Extensive benchmarks and finite population Monte Carlo simulations demonstrate large computational speedups at 1,000 domains (1,263x for Fay-Herriot and 67x for spatial Fay-Herriot relative to sae, 217x for spatial Beta relative to Stan-based MCMC) and reductions of more than 99% in allocated memory, without loss of inferential precision.

1 Introduction

In modern evidence-based governance, policy interventions are increasingly decentralized. Monitoring the United Nations Sustainable Development Goals (SDGs), allocating municipal fiscal transfers, targeting social safety net subsidies, and responding to localized disease outbreaks all require dependable statistical indicators at highly disaggregated geographic administrative levels (Rao and Molina, 2015; Pfeffermann, 2013). However, national statistical offices (NSOs) face structural sample size limitations. Large-scale household surveys—such as the American Community Survey (ACS) in the United States, the European Union Statistics on Income and Living Conditions (EU-SILC), and the National Socioeconomic Survey (Susenas) in Indonesia—are powered to yield acceptable sampling errors only at national or broad regional levels. When partitioned into sub-domains (e.g., municipalities, districts, or health zones), domain-specific sample sizes become sparse or zero, causing direct survey estimators to suffer from high sampling variance and unacceptable coefficients of variation (Pratesi, 2016; Rao and Molina, 2015).

Small area estimation (SAE) overcomes these limitations by "borrowing strength'' across domains and time periods via linking models that combine survey responses with auxiliary administrative registers, census aggregates, or geospatial layers (Rao and Molina, 2015). The cornerstone of this literature remains the area-level Fay-Herriot (FH) model (Fay and Herriot, 1979), extended to account for spatial dependencies via Simultaneous Autoregressive (SAR) structures (Petrucci and Salvati, 2006; Pratesi and Salvati, 2008) and temporal dynamics via panel autoregressive processes (Marhuenda et al., 2013). In the presence of micro-level auxiliary data, the nested error regression model of Battese et al. (1988) (BHF) provides unit-level empirical best linear unbiased predictors (EBLUP).

Despite these theoretical advances, operationalizing small area models within production environments in NSOs has been hindered by three structural bottlenecks: %

1. **Frequentist Computational and Memory Bottlenecks**: Existing implementations of spatial and spatio-temporal Fay-Herriot models rely on interpreted languages. Estimating model variance components via Restricted Maximum Likelihood (REML) requires repeated inversions

Feature / Package	fastsae	sae	emdi	tipsae	mcmcsae	SUMMER
Computational Engine	C++17 / INLA	Pure R	R / nlme	Stan HMC	C++ / MCMC	INLA
Frequentist EBLUP (FH/SFH)	✓	✓	✓	×	×	×
Spatio-Temporal Panel EBLUP	✓	✓	×	×	×	×
Unit-Level BHF Model	✓	✓	✓	×	×	×
OpenMP Parallel Bootstrap	✓	×	×	×	×	×
Non-Gaussian Likelihoods	6 Families	×	Box-Cox	Beta only	4 Families	3 Families
Deterministic Bayesian HB (INLA)	✓	×	×	×	×	✓
Survey Dispersion Matching	✓	×	×	✓	×	×
Modern Spatial Priors (BYM2/Leroux)	✓	×	×	Besag	Besag	BYM2
Two-Stage Hierarchical Calibration	✓	×	×	×	×	×
Official Statistics Reliability Flags	✓	×	×	×	×	×
Automated Production Mapping	✓	×	✓	×	×	✓
Multi-Sheet Dissemination Export	✓	×	✓	×	×	×

Table 1: Comparative landscape of Small Area Estimation packages in the R ecosystem.

of $D \times D$ or $DT \times DT$ covariance matrices. When evaluating Mean Squared Error (MSE) through parametric or non-parametric bootstrap resampling (González-Manteiga et al., 2008), runtimes escalate to hours. Crucially, memory allocation for dense covariance arrays grows quadratically ($O(D^2)$ to $O(D^2T^2)$), triggering out-of-memory (OOM) crashes on large national grids ($D \geq 500$).

2. **Distributional Inflexibility and MCMC Friction in Bayesian SAE**: Real-world welfare indicators are rarely Gaussian. Rates and proportions (poverty headcounts, unemployment, stunting) are strictly bounded in $(0, 1)$, while expenditures exhibit severe positive skewness (Gamma), and demographic events represent discrete counts (Poisson, Negative Binomial) (Janicki, 2020; Fabrizi et al., 2011; Boubeta et al., 2016). Under non-Gaussian likelihoods the linear EBLUP is no longer appropriate; the alternatives are the empirical best predictor (EBP) of a generalized linear mixed model (Boubeta et al., 2016) and hierarchical Bayes (HB) estimation, which also propagates the uncertainty about the variance parameters (Rao and Molina, 2015). Currently, Bayesian SAE software relies on Markov Chain Monte Carlo (MCMC) sampling via **Stan** or **JAGS** (De Nicolò and Gardini, 2024). MCMC sampling requires lengthy runtimes, complex convergence diagnostics ($\hat{R}$, trace plots), and often suffers from divergent transitions on complex spatial graphs (Simpson et al., 2017).
3. **The Institutional Benchmarking and Dissemination Gap**: In official statistics, model-based small area estimates must numerically harmonize with published higher-level totals (e.g., provincial and national direct estimates) to preserve institutional credibility and prevent contradictory policy decisions (You and Rao, 2002; Wang et al., 2008; Stefan and Hidiroglou, 2021). Existing packages rarely offer two-stage hierarchical benchmarking. Furthermore, software ecosystems typically end at model fitting, omitting essential dissemination steps such as automated reliability tagging based on official Relative Standard Error (RSE) thresholds, model concordance evaluation, and publication-ready spatial choropleth mapping.

As summarized in Table 1, the current software landscape is compartmentalized. Foundational toolkits such as **sae** (Molina and Marhuenda, 2015) are implemented in interpreted R, lack multithreading, and are restricted to Gaussian linear frameworks. Packages such as **emdi** (Kreutzmann et al., 2019) offer visualizations but lack spatio-temporal dynamics and non-Gaussian spatial Bayesian likelihoods. Specialized packages such as **tipsae** (De Nicolò and Gardini, 2024) implement Beta regression but depend on MCMC runtimes that scale poorly for panel graphs. Toolkits such as **mcmcsae** (Boonstra, 2022) remain simulation-based, while **SUMMER** (Li et al., 2019) is tailored to demographic survey mortality rather than general survey design-variance matching.

To resolve these limitations, this paper introduces **fastsae**. Specifically, this paper makes four concrete, testable contributions: %

1. **Compiled C++17 Core with OpenMP Acceleration**: We engineer an optimized linear algebra core in C++17 via **RcppArmadillo** that speeds up classical, spatial, and spatio-temporal Fay-Herriot estimation with analytical MSE by $1,263\times$, $67\times$, and $24\times$ relative to **sae** (and the classical model by $42,874\times$ relative to **emdi**) at $D = 1,000$, while reducing allocated memory from gigabytes to a few megabytes. Sparse Cholesky factorizations and structured Woodbury identities avoid every dense $D \times D$ (or $DT \times DT$) inverse.
2. **Unified Deterministic Bayesian Hierarchical Modeling via INLA with Survey Variance Matching**: We formulate a Hierarchical Bayes interface (`hb_area()`) spanning six probability distributions (Gaussian, Beta, Gamma, Binomial, Poisson, Negative Binomial) powered by Integrated Nested Laplace Approximations (Rue et al., 2009), with an optional MCMC engine

in Python (JAX/NumPyro) selected through the same interface (`engine = inla_engine()` or `engine = jax_engine()`). It fixes the likelihood dispersion at the survey sampling variances—the Beta precision of Janicki (2020), evaluated at the model mean in a second stage, and Gamma CV matching—with default priors chosen by simulation (Table 2), achieving 117× to 530× speedups over Stan-based MCMC at the largest benchmarked sizes while maintaining near-perfect inferential concordance ($r = 0.9993$).

3. **Two-Stage Hierarchical Benchmarking Engine**: We implement a modular calibration engine (`benchmark_sae()`) that simultaneously enforces consistency across multi-level administrative hierarchies (Level 0 National, Level 1 Provincial, Level 2 District) via ratio, difference, variance-weighted optimal, and logit-scale adjustments.
4. **End-to-End Official Statistics Dissemination Suite**: We provide turnkey post-estimation workflows, including automated reliability tagging (`add_reliability_flags()`) based on official NSO thresholds, model concordance diagnostics (`compare_sae()`), publication-ready choropleth mapping (`map_sae()`), and multi-sheet spreadsheet exports (`export_sae()`).

2 Methodological framework

2.1 Frequentist small area models

Classical Fay-Herriot model

Let m denote the number of small areas indexed by $i = 1, \ldots, m$. The classical area-level model (Fay and Herriot, 1979) is defined by: %

$$y_i = \boldsymbol{x}_i^\top \boldsymbol{\beta} + u_i + e_i, \quad u_i \overset{\text{iid}}{\sim} \mathcal{N}(0, \sigma_u^2), \quad e_i \overset{\text{ind}}{\sim} \mathcal{N}(0, D_i), \tag{1}$$

% where y_i is the design-based direct estimator of the target indicator θ_i, $\boldsymbol{x}_i$ is a $p \times 1$ vector of auxiliary covariates, D_i is the known design variance, and u_i is the area random effect. The Empirical Best Linear Unbiased Predictor (EBLUP) is given by: %

$$\hat{\theta}_i = \hat{\gamma}_i y_i + (1 - \hat{\gamma}_i) \boldsymbol{x}_i^\top \hat{\boldsymbol{\beta}}, \quad \hat{\gamma}_i = \frac{\hat{\sigma}_u^2}{\hat{\sigma}_u^2 + D_i}, \tag{2}$$

% where $\hat{\boldsymbol{\beta}} = (\boldsymbol{X}^\top \hat{\boldsymbol{V}}^{-1} \boldsymbol{X})^{-1} \boldsymbol{X}^\top \hat{\boldsymbol{V}}^{-1} \boldsymbol{y}$ with $\boldsymbol{V} = \text{diag}(\sigma_u^2 + D_1, \ldots, \sigma_u^2 + D_m)$. Mean squared error is evaluated via the second-order Prasad-Rao approximation (Prasad and Rao, 1990): %

$$\text{MSE}(\hat{\theta}_i) \approx g_{1i}(\hat{\sigma}_u^2) + g_{2i}(\hat{\sigma}_u^2) + 2 g_{3i}(\hat{\sigma}_u^2). \tag{3}$$

Spatial Fay-Herriot model

To accommodate spatial dependence across contiguous domains, Petrucci and Salvati (2006) and Pratesi and Salvati (2008) modeled the area random effects $\boldsymbol{u} = (u_1, \ldots, u_m)^\top$ via a Simultaneous Autoregressive (SAR) process: %

$$\boldsymbol{u} = \rho \boldsymbol{W} \boldsymbol{u} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}_m(\boldsymbol{0}, \sigma_1^2 \boldsymbol{I}_m), \tag{4}$$

% where ρ is the spatial autoregressive parameter and $\boldsymbol{W}$ is a row-standardized spatial proximity matrix. The marginal covariance matrix is $\boldsymbol{V}(\sigma_1^2, \rho) = \boldsymbol{G}(\sigma_1^2, \rho) + \boldsymbol{D}$, where $\boldsymbol{G}(\sigma_1^2, \rho) = \sigma_1^2 [(\boldsymbol{I}_m - \rho \boldsymbol{W})^\top (\boldsymbol{I}_m - \rho \boldsymbol{W})]^{-1}$. The spatial EBLUP is: %

$$\hat{\boldsymbol{\theta}} = \boldsymbol{X} \hat{\boldsymbol{\beta}} + \hat{\boldsymbol{G}} \hat{\boldsymbol{V}}^{-1} (\boldsymbol{y} - \boldsymbol{X} \hat{\boldsymbol{\beta}}). \tag{5}$$

% The SAR process is defined on all domains of $\boldsymbol{W}$, including unsampled ones, which are predicted by spatial kriging. Variance components (σ_1^2, ρ) are estimated by REML or ML Fisher scoring with $|\rho| \leq 0.999$, falling back to a constrained (projected) Fisher scoring when the classical iteration ends at a negative variance. All quantities are obtained from sparse Cholesky factorizations of $\boldsymbol{A} = (\boldsymbol{I} - \rho \boldsymbol{W})^\top (\boldsymbol{I} - \rho \boldsymbol{W})$ and $\boldsymbol{A} + \sigma_1^2 \boldsymbol{E}^\top \boldsymbol{D}^{-1} \boldsymbol{E}$ through the Woodbury identity, so no dense $m \times m$ inverse is formed. Mean squared error uses the same second-order approximation $g_1 + g_2 + 2g_3 - g_4$ as **sae** (Molina and Marhuenda, 2015), or parametric and non-parametric bootstrap.

Spatio-temporal Fay-Herriot model

When small area data are collected across longitudinal waves $t = 1, \ldots, T$, borrowing strength across space and time yields substantial efficiency gains (Marhuenda et al., 2013): %

$$y_{it} = \boldsymbol{x}_{it}^{\top}\boldsymbol{\beta} + u_{1,it} + u_{2,it} + e_{it}, \tag{6}$$

% where $\boldsymbol{u}_1 = \rho_1 \boldsymbol{W}\boldsymbol{u}_1 + \boldsymbol{\epsilon}_1$ is a spatial SAR effect that is constant over time within each area ($u_{1,it} = u_{1,i}$) and $u_{2,it} = \rho_2 u_{2,i,t-1} + \epsilon_{2,it}$ follows an AR(1) temporal process with $|\rho_2| < 1$, as in Marhuenda et al. (2013). In **fastsae**, the covariance $\boldsymbol{V} = \boldsymbol{A} + \boldsymbol{Z}\boldsymbol{V}_{u_1}\boldsymbol{Z}^{\top}$ (with $\boldsymbol{A}$ block diagonal in $T \times T$ blocks) is never formed: REML scores, Fisher information, and the analytical MSE ($g_1 + g_2 + 2g_3 - g_4$) are computed from $T \times T$, $D \times D$, and low-rank $DT \times (D + p)$ pieces via the Woodbury identity. The panel need not be complete: the model is defined on the full $D \times T$ grid, the variance components are estimated by REML from the sampled cells, and missing periods as well as areas that are never sampled are predicted by the best linear unbiased predictor given the sampled data (Saei and Chambers, 2005), with analytical (second-order) and parametric bootstrap MSE for every cell.

Unit-level Battese-Harter-Fuller model

When unit microdata are accessible alongside population auxiliary totals, **fastsae** implements the nested error regression model (Battese et al., 1988): %

$$y_{dj} = \boldsymbol{x}_{dj}^{\top}\boldsymbol{\beta} + u_d + e_{dj}, \quad d = 1, \ldots, D, \quad j = 1, \ldots, n_d, \tag{7}$$

% yielding EBLUP predictors $\hat{\bar{Y}}_d = f_d \bar{y}_d + (1 - f_d)(\bar{\boldsymbol{X}}_{d,r}^{\top}\hat{\boldsymbol{\beta}} + \hat{\gamma}_d(\bar{y}_d - \bar{\boldsymbol{x}}_d^{\top}\hat{\boldsymbol{\beta}}))$, where $f_d = n_d / N_d$.

2.2 Hierarchical Bayes models

Latent Gaussian models

The Bayesian functions formulate small area estimation as a latent Gaussian model (LGM) (Rue et al., 2009). For the area-level model of `hb_area()`, %

$$y_i \mid \mu_i, \psi_i \sim \pi(y_i \mid \mu_i, \psi_i), \quad \eta_i = g(\mu_i) = \boldsymbol{x}_i^{\top}\boldsymbol{\beta} + u_i, \tag{8}$$

% where $g(\cdot)$ is the canonical link of the family (identity, logit or log), ψ_i a dispersion parameter (described below) and $\boldsymbol{u}$ a zero-mean Gaussian Markov random field (GMRF): independent effects $u_i \sim \mathcal{N}(0, \sigma_u^2)$ or one of the spatial structures below, with temporal and space-time terms added for panel data. `hb_unit()` applies the same construction to unit records with a domain effect, and `hb_twofold()` to sub-areas nested in areas, with an area effect v_i and an independent sub-area effect u_{ij}. INLA approximates the posterior marginals of the area parameters μ_i by nested Laplace approximations and numerical integration over the hyperparameters; **fastsae** uses the full Laplace approximation (`strategy = "laplace"`) by default, which is more accurate than the simplified Laplace approximation for non-Gaussian likelihoods with little information per area.

Survey dispersion and variance matching

To preserve the design-based sampling variances D_i of the direct estimates, **fastsae** fixes the dispersion parameter of each area at the value for which the model variance equals D_i: %

1. **Gaussian**: the precision is fixed to the inverse design variance, $\tau_i = 1/D_i$, which gives the Fay-Herriot model.
2. **Beta**: for proportions $y_i \in (0, 1)$ with logit link, $\mathrm{Var}(y_i) = \mu_i(1 - \mu_i)/(1 + \phi_i)$. Equating this to D_i gives the precision of Janicki (2020),

$$\phi_i = \max\left(\frac{\mu_i(1 - \mu_i)}{D_i} - 1,\, 1\right). \tag{9}$$

   It depends on the unknown μ_i. Replacing μ_i by the direct estimate y_i makes the model variance $D_i\mu_i(1 - \mu_i)/\{y_i(1 - y_i)\}$, too small wherever y_i falls far from μ_i, and in a simulation with proportions near 0.08 the 95% intervals covered only 81–84% of the true values. **fastsae** therefore fits the model twice and evaluates (9) at the posterior means of the first fit, which restored a coverage of 90–94%. With known sample sizes n_i instead of D_i, $\phi_i = n_i - 1$.

Parameter	Default prior
Coefficients $\boldsymbol{\beta}$	intercept flat; others $\mathcal{N}(0, 1000)$
Random-effect SDs (area, sub-area, time, space-time; σ_e of `hb_unit()`)	half-normal(s) with $s = \mathrm{sd}(\boldsymbol{y})$ for Gaussian models and $s = 1$ on the link scale otherwise
BYM2 mixing parameter ϕ	PC prior with $P(\phi < 0.5) = 0.5$ (Riebler et al., 2016)
Leroux λ; SAR and SLM ρ	uniform on the admissible range
AR(1) correlation ρ	$\mathcal{N}(0, 1/0.15)$ on $\log\{(1+\rho)/(1-\rho)\}$
Gamma shape, Negative Binomial size (no D_i)	half-normal(10)
Beta precision (no D_i, no n_i)	$\phi \sim$ Exponential(0.1)
Gaussian residual SD (no D_i)	half-normal(s)

Table 2: Default priors of `hb_area()`, `hb_unit()` and `hb_twofold()`. All intrinsic GMRFs are scaled (Sørbye and Rue, 2014), so σ is a typical marginal standard deviation for every structure. Any prior can be replaced through `inla_engine()`.

3. **Gamma**: for positive skewed indicators with log link, equating $\mathrm{Var}(y_i) = \mu_i^2/s_i$ to D_i gives the shape $s_i = y_i^2/D_i = 1/\mathrm{CV}_i^2$.
4. **Negative Binomial**: equating $\mathrm{Var}(y_i) = \mu_i + \mu_i^2/k_i$ to D_i gives the size $k_i = \hat{\mu}_i^2/(D_i - \hat{\mu}_i)$, again at the posterior means $\hat{\mu}_i$ of a first fit.
5. **Binomial and Poisson**: the domain sample sizes n_i are the numbers of trials, and population exposures E_i enter as the offset $\log(E_i)$.

% Without sampling variances, a single dispersion parameter is estimated. With one direct estimate per area it trades off against the area effect, and its prior then determines the intervals (Table 2); `hb_area()` warns in this case.

Spatial priors and space-time interactions

The area effect $\boldsymbol{u}$ supports five graph-based structures: the intrinsic CAR (ICAR) model of Besag et al. (1991), the BYM model with independent and ICAR components, the scaled BYM2 model of Riebler et al. (2016), $u_i = \sigma_u(\sqrt{1-\phi}\, v_i + \sqrt{\phi}\, u_i^*)$ with u^* a scaled ICAR field, the Leroux CAR with precision $\boldsymbol{Q} = \tau[(1-\lambda)\boldsymbol{I} + \lambda(\mathrm{diag}(\boldsymbol{d}) - \boldsymbol{W})]$ (Leroux et al., 2000), and simultaneous autoregressive models, either for the area effect, $\boldsymbol{u} = (\boldsymbol{I} - \rho\boldsymbol{W})^{-1}\boldsymbol{\varepsilon}$ as in the spatial Fay-Herriot model, or as a spatial lag model (SLM). For panel observations across $T \geq 2$, temporal trends support random walks (RW1, RW2), AR(1) and independent effects. Space-time interactions can be specified as domain-specific trends, a separable spatial field with AR(1) dynamics, or according to the four Knorr-Held (2000) interaction types ($\boldsymbol{Q}_{st} = \boldsymbol{Q}_s \otimes \boldsymbol{Q}_t$ for type IV).

Default prior distributions

With few areas, or one observation per area, the data hardly inform the variance and dispersion parameters, and their priors decide how strongly the estimates are shrunk and how wide the intervals are. Table 2 lists the defaults, which are shared by the INLA and JAX engines so that both fit the same model.

The half-normal prior is the weakly informative choice of Gelman (2006) for hierarchical standard deviations. Its scale is tied to the response: a penalized complexity (PC) prior $P(\sigma > U) = \alpha$ (Simpson et al., 2017) is a good default when U is chosen for the problem, but the common $U = 1$, $\alpha = 0.01$ depends on the unit of y. In a Fay-Herriot simulation with 30 areas the 95% intervals under PC$(1, 0.01)$ covered 94%, 85% and 38% of the true means for the same data expressed in three units, while the scaled half-normal prior covered as the EBLUP intervals (93–97% for 15 to 60 areas, EBLUP 93–96%), and PC priors with $U = \mathrm{sd}(\boldsymbol{y})$ still covered only 85–94% because they put much mass near $\sigma = 0$. On the logit and log scales, half-normal(1) covered at least as well as half-normal(2.5), the default of **tipsae**, and estimated the area SD with less bias. For the dispersion parameters estimated without D_i, INLA's own defaults put the Gamma shape near 80 and the Negative Binomial size near 110–180 whatever the truth, and intervals for overdispersed data then covered 52–92% of the true means; the half-normal(10) prior covered 90–99%, at the cost of too wide intervals when the sampling noise is small, which is why D_i should be supplied whenever it is known. The simulations and the code behind them are documented in the package vignette `vignette("hb-priors")`.

2.3 Two-stage hierarchical benchmarking calibration

In official statistics, model-based small area estimates $\hat{\theta}_{di}$ for domain i in province/region g must aggregate to officially published higher-level targets (Rao and Molina, 2015; Stefan and Hidiroglou, 2021). Let w_{di} denote domain population weights such that $W_g = \sum_{i \in g} w_{di}$ is the provincial total weight, and $W_{\text{nat}} = \sum_g W_g$ is the national total weight. Let T_g denote the provincial target and T_{nat} denote the national aggregate target.

fastsae implements a two-stage hierarchical benchmarking calibration: %

1. **Stage 1: Harmonization of Provincial Targets to National Benchmark**: When initial provincial targets T_g (or initial aggregated model estimates $\hat{\tilde{\theta}}_g = \sum_{i \in g}(w_{di}/W_g)\hat{\theta}_{di}$) do not sum to the national target T_{nat}, calibrated provincial targets $\tilde{T}_g$ are computed to satisfy $\sum_g (W_g/W_{\text{nat}})\tilde{T}_g = T_{\text{nat}}$.
2. **Stage 2: Domain-Level Calibration within Provinces**: Domain estimates within province g are calibrated to satisfy $\sum_{i \in g}(w_{di}/W_g)\hat{\theta}^{\text{BM}}_{di} = \tilde{T}_g$.

% Across both stages, **fastsae** supports four calibration mechanisms: %

- **Ratio Benchmarking** (You and Rao, 2002): $\hat{\theta}^{\text{BM}}_{di} = \hat{\theta}_{di}\left(\tilde{T}_g / \sum_{j \in g} \tilde{w}_{dj}\hat{\theta}_{dj}\right)$.
- **Difference Benchmarking** (Rao and Molina, 2015): $\hat{\theta}^{\text{BM}}_{di} = \hat{\theta}_{di} + \left(\tilde{T}_g - \sum_{j \in g} \tilde{w}_{dj}\hat{\theta}_{dj}\right)$.
- **Variance-Weighted Optimal Benchmarking** (Datta et al., 2011; Bell et al., 2013): Adjustments minimize $\sum_i (\hat{\theta}^{\text{BM}}_{di} - \hat{\theta}_{di})^2/\text{MSE}(\hat{\theta}_{di})$ subject to the benchmark constraint, which allocates them in proportion to $\tilde{w}_{di}\,\text{MSE}(\hat{\theta}_{di})$:

$$\hat{\theta}^{\text{BM}}_{di} = \hat{\theta}_{di} + \tilde{w}_{di}\text{MSE}(\hat{\theta}_{di})\left(\sum_{j \in g} \tilde{w}^2_{dj}\text{MSE}(\hat{\theta}_{dj})\right)^{-1}\left(\tilde{T}_g - \sum_{j \in g} \tilde{w}_{dj}\hat{\theta}_{dj}\right). \tag{10}$$

- **Logit-Scale Benchmarking**: For bounded proportions in $(0,1)$, linear adjustments can push estimates outside $[0,1]$. **fastsae** solves for an additive shift α_g on the logit scale:

$$\sum_{i \in g} \tilde{w}_{di}\,\text{logit}^{-1}\left(\text{logit}(\hat{\theta}_{di}) + \alpha_g\right) = \tilde{T}_g, \tag{11}$$

  via robust one-dimensional root-finding, guaranteeing $\hat{\theta}^{\text{BM}}_{di} \in (0,1)$.

2.4 Official statistics reliability classification

NSOs classify domain estimates into publication tiers based on the Relative Standard Error ($\text{RSE} = \frac{\sqrt{\text{MSE}(\hat{\theta}_i)}}{|\hat{\theta}_i|} \times 100\%$). **fastsae** provides `add_reliability_flags()` implementing institutional thresholding: %

$$\text{Tier}_i = \begin{cases} \text{Reliable (Direct Publish)}, & \text{RSE}_i \le c_1, \\ \text{Use with Caution (Accompanying Warning)}, & c_1 < \text{RSE}_i \le c_2, \\ \text{Unreliable (Suppress or Aggregate)}, & \text{RSE}_i > c_2, \end{cases} \tag{12}$$

% where the default thresholds $(c_1, c_2) = (25\%, 50\%)$ follow the criteria of BPS-Statistics Indonesia (with $\text{RSE} \le c_1$ reliable and $\text{RSE} > c_2$ unreliable); other agencies' cut-offs can be supplied.

3 Software design and implementation

The architecture of **fastsae** unites compiled linear algebra with high-level Bayesian GMRF modeling and production reporting. Figure 1 outlines the structural flow.

The frequentist engines (`eblup_fh()`, `eblup_sfh()`, `eblup_stfh()`, `eblup_bhf()`, `eblup_twofold()`) never form a dense inverse of the marginal covariance: the spatial model works with sparse Cholesky factors (**RcppEigen**), the spatio-temporal and two-fold models with block-structured Woodbury identities, and the unit-level model with domain-level sufficient statistics, profiling $\boldsymbol{\beta}$ and σ^2_e out of the (restricted) likelihood. The bootstrap MSE loops run in parallel with OpenMP (`#pragma omp parallel for`); random numbers are drawn sequentially from R's generator beforehand, so results depend only on the seed and not on the number of threads. The Bayesian functions (`hb_area()`, `hb_unit()`,

fastsae High-Level Software Architecture

1. User-Facing Production API (R)		
Frequentist Modeling	**Bayesian HB Engine**	**Production Workflow**
`eblup_fh()`	`hb_area(...,family = "beta")`	`benchmark_sae()` (Two-Stage)
`eblup_sfh()`	`hb_area(...,family = "gamma")`	`compare_sae()`
`eblup_stfh()`	`hb_area(...,family = "poisson")`	`add_reliability_flags()`
`eblup_bhf()`	`hb_area(...,spatial = "bym2")`	`map_sae()` / `export_sae()`
2. Core Computational Backends		
Compiled C++17 Core	**INLA Deterministic Engine**	**Spatial Linear Algebra**
RcppArmadillo Matrix Algebra	Sparse Cholesky Factorization	Row-normalized Proximity
OpenMP Threaded Bootstrap	Full Laplace Approximation	Scaled Priors & GMRF Graphs
In-Place Fisher-Scoring Updates	Survey Dispersion Injections	Spectral Eigenvalue Bounds

Figure 1: Structural architecture and functional components of fastsae.

D	Execution Time: Median (seconds)			Memory Allocated (MB)		
	fastsae	sae	emdi	fastsae	sae	emdi
Fay-Herriot EBLUP (Classical)						
30	0.00055	0.00155	0.01088	0.339	0.153	4.18
100	0.00064	0.00344	0.05051	0.018	0.918	37.34
500	0.00107	0.19673	5.86366	0.085	18.239	880.31
1000	0.00119	1.50363	51.0224	0.503	71.977	3,773.60
Spatial Fay-Herriot (SEBLUP)						
30	0.00588	0.00410	0.01071	0.009	1.373	4.18
100	0.01388	0.07054	0.05299	0.021	17.976	37.34
500	0.21310	7.11793	3.13662	0.101	776.407	949.86
1000	0.99977	67.1518	51.0593	0.203	1,920.01	3,772.57
Spatio-Temporal Fay-Herriot (STEBLUP, $T = 5$)						
30	0.14272	0.11856	–	0.138	43.801	–
100	0.21415	17.9425	–	0.316	2,551.10	–
500	10.0430	253.343	–	1.633	8,124.37	–
1000	75.3340	1,786.49	–	3.293	32,478.3	–

Table 3: Frequentist runtime and memory allocation (point estimates and analytical MSE) across domain sample sizes (D). At $D = 1,000$, fastsae is $1,263\times$ faster than sae for FH, $67\times$ for SFH, and $24\times$ for ST-FH. At $D = 30$ the spatial and spatio-temporal fits are dominated by fixed costs, and sae is slightly faster.

`hb_twofold()`) translate formulas, spatial adjacency matrices $\boldsymbol{W}$, and survey dispersion arguments into **R-INLA** calls. How a model is estimated is specified separately from the model by an engine object: `inla_engine()` holds the INLA strategy, priors, and further `INLA::inla()` arguments, and `jax_engine()` fits `hb_area()` models by NUTS sampling in JAX/NumPyro (CPU or GPU) through reticulate. Standard S3 methods (`print()`, `summary()`, `plot()`, `coef()`, `fitted()`, `residuals()`) are implemented across all model classes.

4 Computational benchmarks and simulations

4.1 Frequentist scalability and memory efficiency: fastsae vs. sae and emdi

We evaluated computational speed and memory allocation across domain sizes $D = 30, 50, 100, 250, 500,$ and 1000 using simulated datasets with 5 auxiliary covariates. Times are medians measured with the bench package; memory is the total memory allocated by R during one call (`mem_alloc`), not the peak resident memory. The fastsae rows were measured with fastsae 1.0.0 under R 4.6.1 on an Apple M4 processor (10 cores); the sae and emdi rows come from an earlier run of the same design on different random draws. The scripts are in the `benchmarks/` directory of the package repository.

Table 3 demonstrates the scalability gains of fastsae. At $D = 1,000$, classical FH completes in about

D	Cross-Sectional Beta		Spatial Beta (Besag)		Spatio-Temporal Beta	
	fastsae	tipsae	fastsae	tipsae	fastsae	tipsae
30	1.20 s	2.35 s	1.22 s	25.72 s	1.48 s	540.57 s
50	1.23 s	3.77 s	1.23 s	65.72 s	1.71 s	908.17 s
100	1.26 s	8.24 s	1.25 s	100.02 s	–	–
250	1.32 s	26.28 s	1.33 s	155.90 s	–	–
500	1.40 s	64.21 s	1.46 s	248.82 s	–	–
1000	1.61 s	187.16 s	1.87 s	404.34 s	–	–

Table 4: Median execution time for Bayesian Beta models (spatio-temporal: $T = 5$). Both packages are fitted to the same simulated data. Point estimates of fastsae and tipsae for the cross-sectional model at $D = 1,000$ achieve $r = 0.9993$ concordance.

one millisecond in fastsae compared to 1.5 seconds in sae (1,263× slower) and 51 seconds in emdi (42,874× slower). For spatio-temporal models at $D = 1,000$ ($T = 5$), sae required nearly 30 minutes (1,786 seconds) and allocated 32.5 gigabytes, whereas fastsae finished in 75 seconds allocating 3.3 megabytes. **Operational Takeaway**: fastsae eliminates out-of-memory bottlenecks, enabling routine spatio-temporal estimation across thousands of domains on standard laptop hardware.

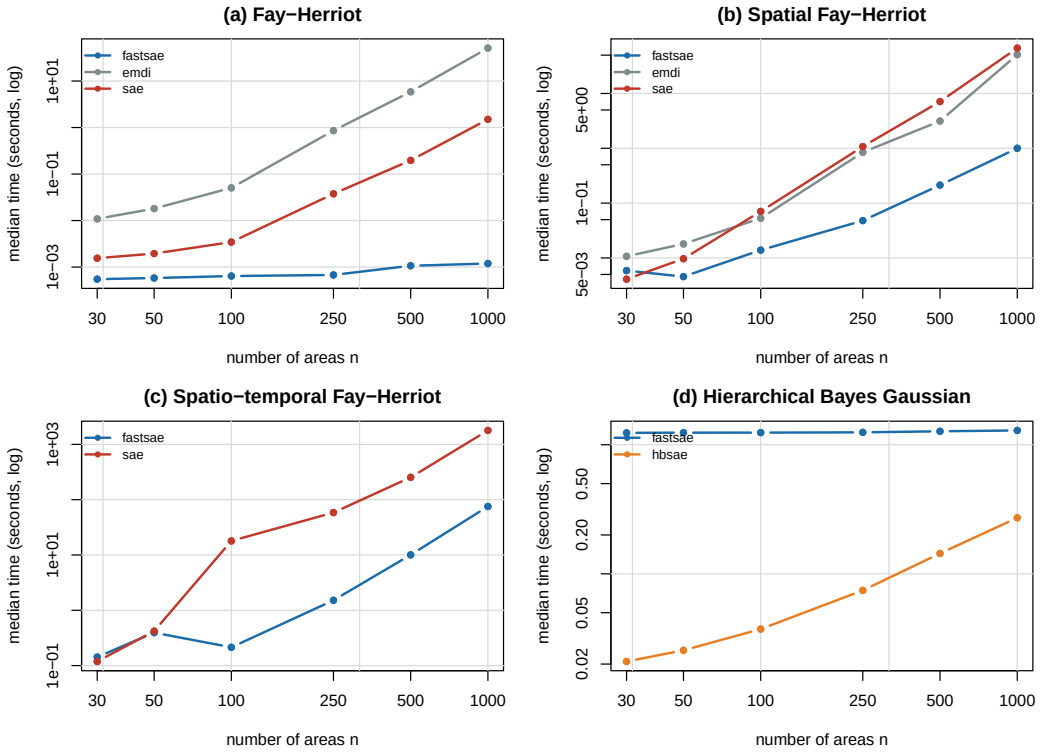


Figure 2: Median runtime scaling across domain sample sizes (D) on log-log scale for classical, spatial, spatio-temporal Fay-Herriot, and Bayesian Gaussian models.

4.2 Bayesian Beta Hierarchical Modeling scalability: fastsae vs. tipsae

For bounded rates, we benchmarked `hb_area(...,family = "beta")` against tipsae (De Nicolò and Gardini, 2024) using Stan (4 chains, 2,000 iterations).

As shown in Table 4, deterministic Laplace approximations in fastsae scale flatly with domain count: fitting a spatial Beta model at $D = 1,000$ requires only 1.87 seconds in fastsae compared to 404 seconds in tipsae (217× speedup). Most of the fastsae time is the fixed start-up cost of the INLA program (about 1.2 seconds even at $D = 30$). In spatio-temporal panel models ($D = 50, T = 5$), tipsae required 15.1 minutes (908 seconds), whereas fastsae completed in 1.71 seconds (530× speedup). Despite this speed differential, inferential concordance on $D = 1,000$ yields Pearson correlation $r = 0.9993$, Spearman $\rho = 0.9992$, and Mean Absolute Difference MAD $= 0.0197$. **Operational**

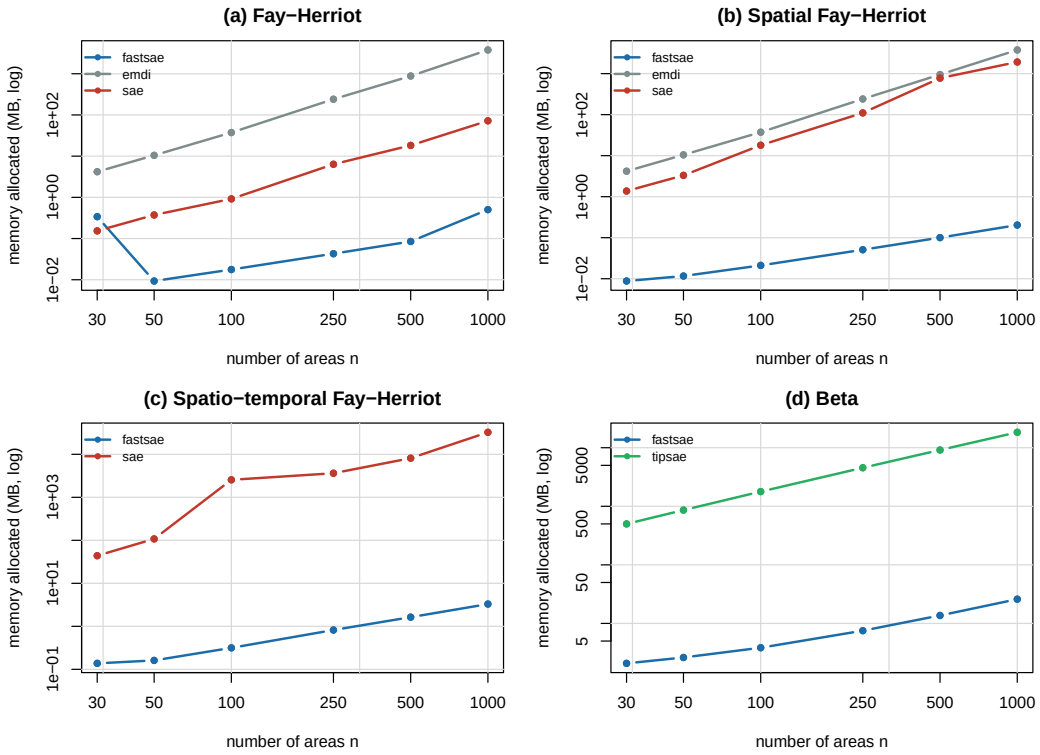


Figure 3: Memory allocated per call (MB, `bench::mark()$mem_alloc`) across domain counts (D) on log-log scale. **fastsae** avoids quadratic $O(D^2T^2)$ allocation.

Model	D	INLA	JAX (CPU)	JAX (Apple GPU)	**tipsae**
Gaussian	100	1.4 s	6.4 s	129 s	–
	500	1.4 s	45 s	155 s	–
	1,000	1.5 s	94 s	205 s	–
	2,500	2.5 s	713 s	474 s	–
Beta	100	1.2 s	6.0 s	110 s	107 s
	500	1.4 s	36 s	136 s	268 s
	1,000	1.8 s	78 s	201 s	401 s
	2,500	3.2 s	538 s	393 s	946 s

Table 5: Warm runtime of `hb_area()` with an ICAR area effect by estimation engine on an Apple M4. JAX: NumPyro NUTS, 2 chains × 1,000 draws after 500 warmup. **tipsae**: Stan defaults, Beta model only.

Takeaway: Deterministic Laplace approximation in **fastsae** achieves inferential equivalence with fully converged Hamiltonian MCMC in seconds rather than minutes or hours.

4.3 Estimation engines: INLA versus MCMC on CPU and GPU

The same `hb_area()` model can be estimated by INLA (`inla_engine()`) or by full MCMC with the No-U-Turn sampler of NumPyro on JAX (`jax_engine()`), on the CPU or on a GPU. We compared the engines on spatial models with a Besag (ICAR) area effect on a regular grid with queen adjacency, for a Gaussian Fay-Herriot model and a Beta model; for the Beta model, **tipsae** was fitted with its ICAR spatial error. Each engine ran in its own R process on an Apple M4 (10 cores, 16 GB), JAX with 2 chains of 1,000 draws after 500 warmup, and each fit was run twice on the same data; Table 5 reports the second (warm) run.

INLA is the fastest engine at every size: at $D = 2,500$ it fits the spatial Beta model in 3.2 s, 296 times faster than **tipsae**. Among the MCMC options, JAX on the CPU is 2 to 18 times faster than Stan, with the gap narrowing as D grows, because the JAX engine represents the ICAR effect through the eigenvectors of the $D \times D$ graph Laplacian ($O(D^2)$ work per gradient). The Apple GPU carries a fixed cost of about two minutes per fit, since each step of the NUTS tree-building loop synchronises the GPU

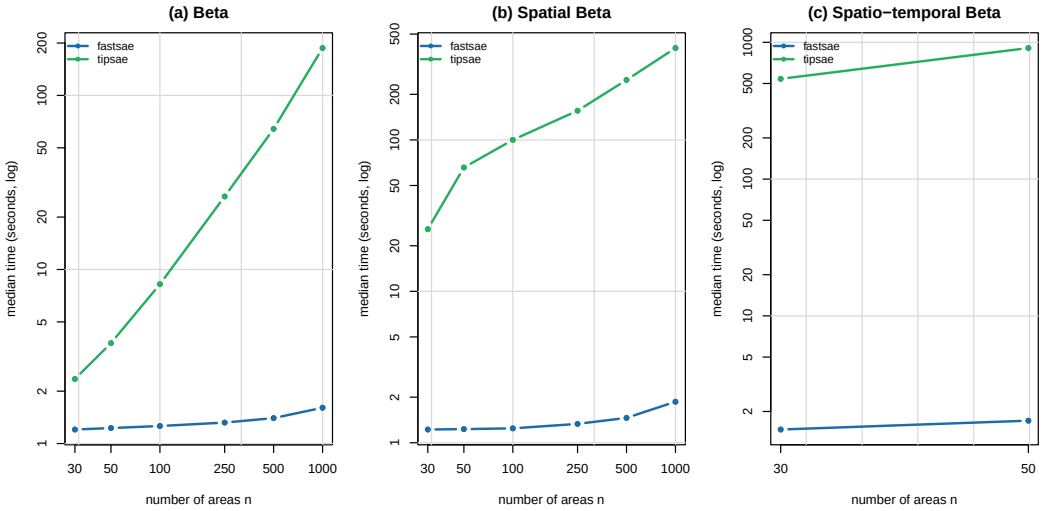


Figure 4: Runtime scaling across domain sample sizes (D) for cross-sectional, spatial, and spatio-temporal Bayesian Beta models.

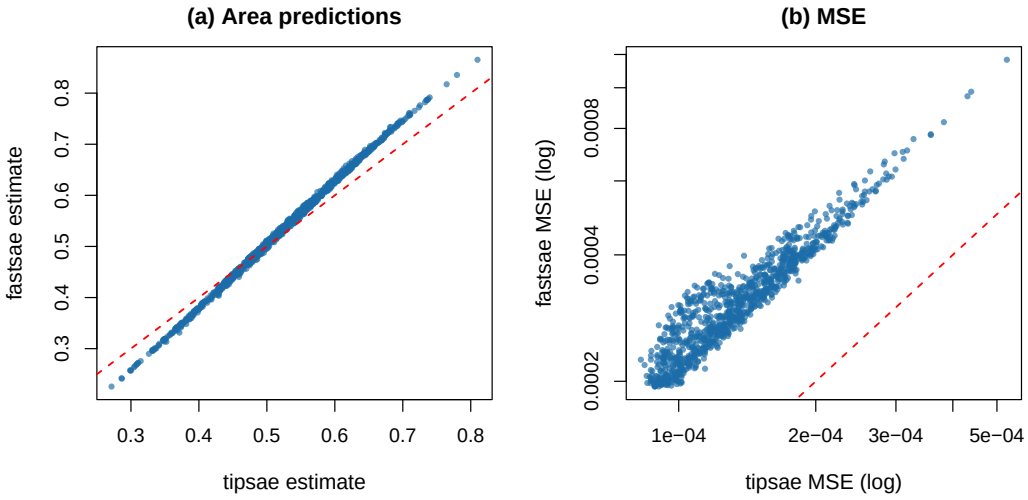


Figure 5: Numerical equivalence and inferential concordance between **fastsae** (INLA) and **tipsae** (Stan HMC) at $D = 1{,}000$ ($r = 0.9993$).

Estimator	Emp. RMSE	RRMSE (%)	MARB (%)	95% CrI Cov.	CrI Width	Runtime
Direct Sample Estimator	0.0719	27.1%	21.7%	–	–	–
fastsae (Spatial Besag)	0.0489	18.9%	15.8%	93.4%	**0.194**	**8.5 s**
tipsae (Spatial Besag)	**0.0487**	**18.7%**	**15.7%**	**93.7%**	0.202	40.0 s

Table 6: Finite population Monte Carlo simulation results ($D = 500$, $R = 25$), averaged over areas. Runtimes per fit with five replicates running in parallel on an Apple M4.

and the CPU, but it overtakes the CPU at $D = 2{,}500$ for both models. Running NumPyro's NUTS on Apple GPUs required two workarounds in fastsae, because **jax-metal** cannot compile the `popcnt` bit count or `lax.cond` statements with constant outputs; both are replaced by equivalent operations when `device = "metal"`. The engines agree closely: the posterior means of JAX on either device correlate with INLA at $r \geq 0.9999$, and the largest split $\hat{R}$ over all JAX fits is 1.007 (1.083 for tipsae with its default run length).

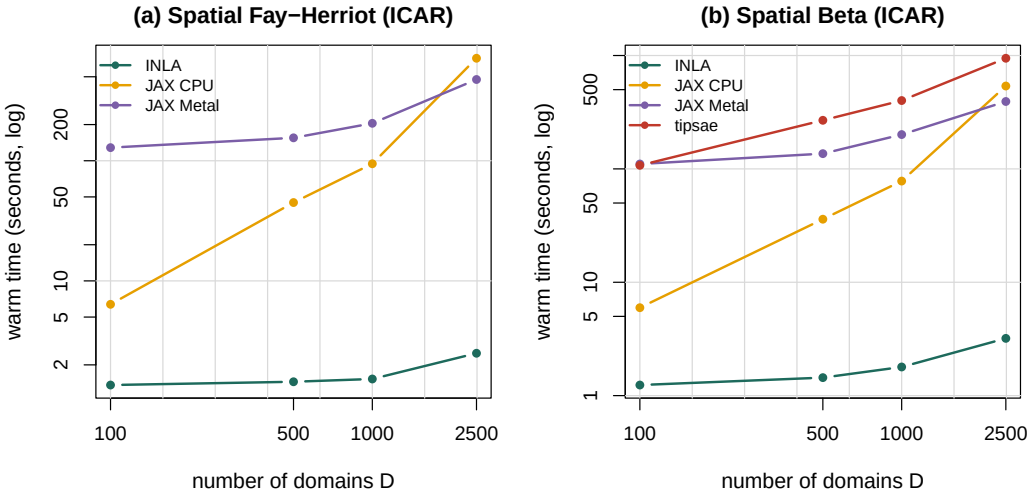


Figure 6: Warm runtime of the `hb_area()` estimation engines for spatial (ICAR) Gaussian and Beta models on an Apple M4 (log scales).

4.4 Monte Carlo finite population validation against ground truth

To assess frequentist design properties, we conducted design-based Monte Carlo simulations under known ground truth at national scale ($D = 500$ small areas on a 20×25 grid, $R = 25$ independent sample draws) using a synthetic finite population of $N = 250{,}000$ individuals with a binary outcome, logistic in two covariates with spatial (ICAR) and independent area effects. Each replicate draws 20 to 50 units per area by simple random sampling and fits a Beta model with an ICAR area effect and the sampling variances of the direct proportions, by `hb_area()` with its default priors and by tipsae (`type_disp = "var"`, one chain of 500 iterations after 150 warmup). The script is `benchmarks/sim_mc_finite_pop_d500.R` in the package repository.

Table 6 shows that `hb_area()` reduces the empirical RMSE from 0.0719 (direct survey estimator) to 0.0489, an empirical efficiency gain (ratio of mean squared errors) of 1.99, and the relative RMSE from 27.1% to 18.9%. The two Bayesian fits are practically equivalent in accuracy (RMSE 0.0489 against 0.0487) and in the coverage of their 95% credible intervals (93.4% and 93.7%), with slightly narrower intervals for fastsae, which needed about a fifth of the runtime per fit. An earlier version of this simulation used a PC prior on the area SD and the plug-in Beta precision $y_i(1 - y_i)/D_i - 1$; its intervals covered only 89.0% of the true proportions, against 93.4% with the two-stage precision and the default priors of Table 2.

5 Applied illustration and production workflow

We demonstrate an end-to-end NSO production workflow using the benchmark dataset `sim_area` ($D = 42$ domains across 6 provincial groups).

5.1 Loading data, spatial weights, and fitting models

```
library("fastsae")
data("sim_area")
sim_area$prov <- rep(paste0("Prov_", 1:6), each = 7)
sim_area$pop <- round(runif(nrow(sim_area), 15000, 45000))

# Construct row-standardized spatial proximity matrix
coords <- cbind(sim_area$x_coord, sim_area$y_coord)
dists <- as.matrix(dist(coords))
W <- (dists < quantile(dists, 0.18)) * 1
diag(W) <- 0
rs <- rowSums(W); rs[rs == 0] <- 1; W_std <- W / rs

# Fit Frequentist and Bayesian Models
fit_fh <- eblup_fh(y_gaussian ~ x1 + x2, vardir = "vardir",
  data = sim_area, method = "REML", print_result = FALSE)
fit_sfh <- eblup_sfh(y_gaussian ~ x1 + x2, vardir = "vardir",
  domain = "area", W = W_std, data = sim_area,
  method = "REML", print_result = FALSE)
fit_beta <- hb_area(y_beta ~ x1 + x2, vardir = "vardir",
  data = sim_area, family = "beta", spatial = "bym2",
  W = W, print_result = FALSE)
```

5.2 Automated model concordance evaluation

Practitioners can immediately compare competitive models via `compare_sae()`:

```
comp <- compare_sae(fit_fh, fit_sfh, names = c("FH", "SFH"))
comp$metrics[c(1, 2, 4, 6, 8, 9), ]
```

```
#>         Group                                      Metric   Value
#> 1     Domains                           Matched Domains      42
#> 2   Agreement                Mean Difference (SFH - FH) -0.0110
#> 4   Agreement             Mean Absolute Difference (MAE)  0.0279
#> 6   Agreement         Lin's Concordance Correlation (CCC)  0.9995
#> 8 Uncertainty % of domains with |Difference| > RMSE of FH    0.0%
#> 9     Ranking             Spearman Rank Correlation (rho)  0.9987
```

5.3 Two-stage hierarchical benchmarking calibration

To harmonize domain estimates with an official national poverty target ($T_{\text{nat}} = 0.50$), we apply two-stage calibration:

```
bm <- benchmark_sae(fit_fh, national_target = 0.50, group = "prov",
  weight = "pop", method = "ratio", type = "mean")
head(bm[, c("domain", "group", "original", "benchmarked", "rel_adjustment_pct")], 4)
```

```
#> === fastsae Small Area Benchmark Calibration ===
#> Method: Ratio (Multiplicative Adjustment)
#> Target Type: Weighted Mean / Rate
#> Total Domains: 4
#>
#> First 6 benchmarked domains (Level 2):
#>   domain  group   original benchmarked rel_adjustment_pct
#> 1      1 Prov_1  2.6646558   0.9317599          -65.03264
#> 2      2 Prov_1 -1.6547233  -0.5786131          -65.03264
#> 3      3 Prov_1  1.2504674   0.4372555          -65.03264
#> 4      4 Prov_1 -0.7988192  -0.2793260          -65.03264
```

The calibrated estimates satisfy $\sum(w_i / \sum w_i)\hat{\theta}_i^{\text{BM}} = 0.50$ exactly.

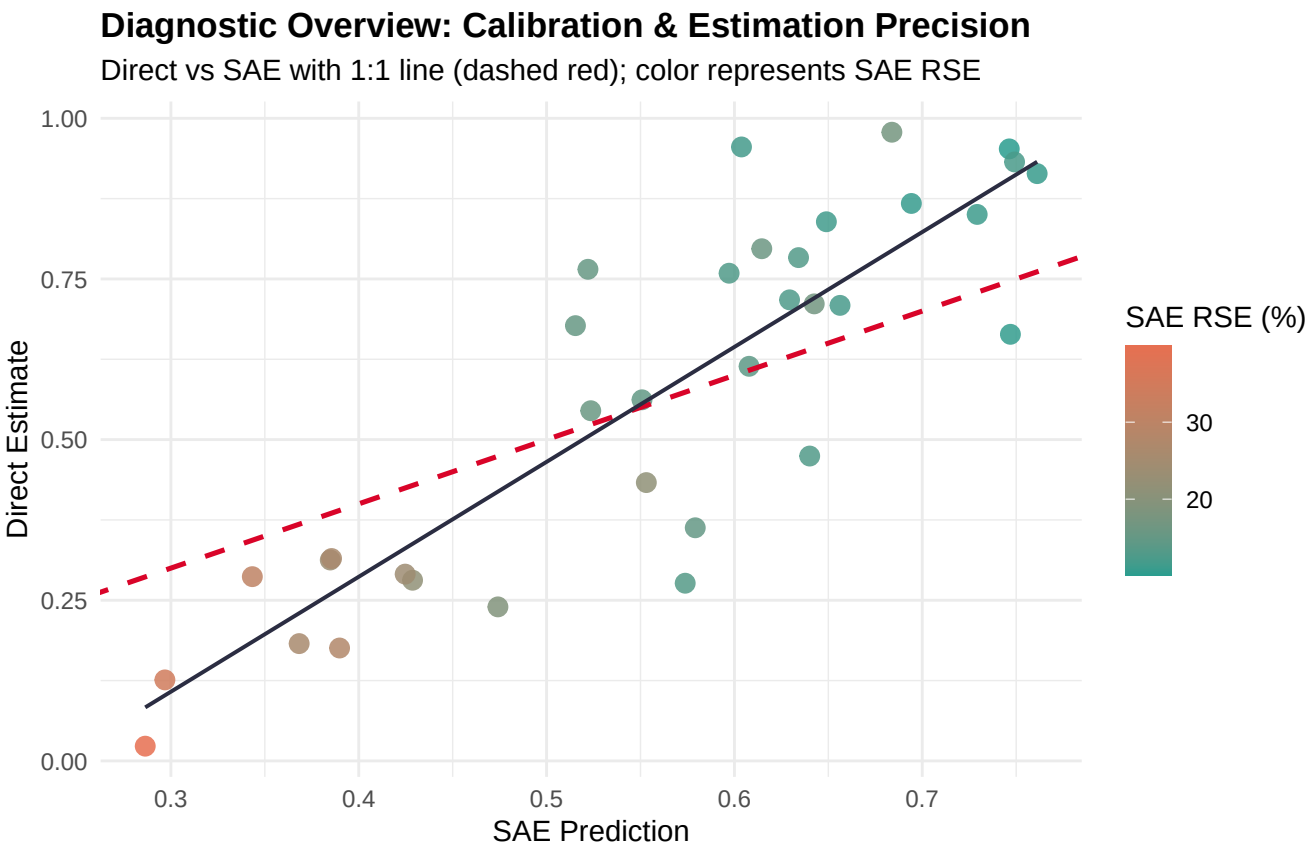


Figure 7: Model diagnostic panels generated via autoplot.

5.4 Official statistics reliability classification

Using `add_reliability_flags()`, domains are classified into official dissemination tiers:

```
flags <- add_reliability_flags(fit_beta)
table(flags$reliability)
```

```
#>
#>               Reliable (RSE <= 25%) Use with caution (25% < RSE <= 50%)
#>                                 35                                     7
#>             Unreliable (RSE > 50%)
#>                                  0
```

5.5 Model diagnostics and visualization

Diagnostics via `diagnose()` and `autoplot()` provide visual quality verification:

```
diag_beta <- diagnose(fit_beta)
autoplot(diag_beta)
```

6 Summary and discussion

Small area estimation in official statistics requires reconciling mathematical rigor with strict computational and operational constraints. **fastsae** resolves these challenges by bridging high-performance compiled linear algebra, deterministic Bayesian approximations, and production-ready dissemination tools.

As synthesized in Table 7, **fastsae** provides substantial efficiency gains across both frequentist and Bayesian paradigms. Future developments will focus on expanding unit-level spatial EBP under non-Gaussian likelihoods, integrating zero-inflated Beta (ZIB) specifications for extreme sparse indicators, and supporting direct spatial polygon streaming.

Dimension	Observed Empirical Evidence	Institutional Implication ("So What")
Frequentist Scalability	$1,263\times$ speedup for FH and $67\times$ for SFH at $D = 1,000$; a few MB allocated for ST-FH (vs 32 GB in **sae**).	Eliminates server memory exhaustion, allowing desktop computers to execute nationwide bootstrap MSE routines.
Bayesian HB via INLA	$117\times$ to $530\times$ speedup over Stan MCMC at the largest benchmarked sizes; point correlation $r = 0.9993$ on $D = 1,000$.	Eliminates manual MCMC chain diagnostics and divergent transition failures in routine production timelines.
Design-Based Ground Truth	$1.99\times$ empirical efficiency gain over direct estimator; 93% empirical coverage of 95% intervals.	Delivers substantial MSE reductions with close to nominal interval coverage under repeated sampling.
Hierarchical Benchmarking	Simultaneous Level 0 and Level 1 target alignment with logit boundary safety.	Guarantees numeric consistency with published aggregates, eliminating public confusion without violating probability bounds.
Turnkey Dissemination	Direct RSE classification, spatial choropleths, and multi-sheet exports.	Collapses multi-script analyst workflows into standardized, reproducible publication pipelines.

Table 7: Synthesis of empirical findings and operational implications of **fastsae**.

Computational details

The **fastsae** results in this paper were obtained using R 4.6.1 with **fastsae** 1.0.0, **R-INLA** 26.8.7, and **ggplot2** 4.0.3 on an Apple M4 processor; the comparison runs used **sae** 1.3, **emdi** 2.2, **tipsae** 1.0, and **hbsae** 1.2. R and packages from CRAN are available at `https://CRAN.R-project.org/`. **R-INLA** is available from `https://www.r-inla.org/`. The **fastsae** package is available on GitHub at `https://github.com/ridsonap/fastsae`.

Acknowledgments

The authors thank the Department of Computational Statistics and Department of Statistics at Politeknik Statistika STIS for institutional support. We also acknowledge the valuable domain feedback provided by colleagues at BPS-Statistics Indonesia.

References

References

G. E. Battese, R. M. Harter, and W. A. Fuller. An error-components model for prediction of county crop areas using survey and satellite data. *Journal of the American Statistical Association*, 83(401):28–36, 1988. doi: 10.1080/01621459.1988.10478561. [p1, 4]

W. R. Bell, G. S. Datta, and M. Ghosh. Benchmarking small area estimators. *Biometrika*, 100(1):189–202, 2013. doi: 10.1093/biomet/ass063. [p6]

J. Besag, J. York, and A. Mollié. Bayesian image restoration, with two applications in spatial statistics. *Annals of the Institute of Statistical Mathematics*, 43(1):1–20, 1991. doi: 10.1007/BF00116466. [p5]

H. J. Boonstra. **mcmcsae***: Markov Chain Monte Carlo Small Area Estimation*, 2022. URL `https://CRAN.R-project.org/package=mcmcsae`. R package version 0.7.3. [p2]

M. Boubeta, M. J. Lombardía, and D. Morales. Empirical best prediction under area-level Poisson mixed models. *TEST*, 25(3):548–569, 2016. doi: 10.1007/s11749-015-0469-8. [p2]

G. S. Datta, M. Ghosh, R. Steorts, and J. Maples. Bayesian benchmarking with applications to small area estimation. *TEST*, 20(3):574–588, 2011. doi: 10.1007/s11749-010-0218-y. [p6]

S. De Nicolò and A. Gardini. The R package **tipsae**: Tools for mapping proportions and indicators on the unit interval. *Journal of Statistical Software*, 108(1):1–36, 2024. doi: 10.18637/jss.v108.i01. [p2, 8]

E. Fabrizi, M. R. Ferrante, S. Pacei, and C. Trivisano. Hierarchical Bayes multivariate estimation of poverty rates based on increasing thresholds for small domains. *Computational Statistics & Data Analysis*, 55(4):1736–1747, 2011. doi: 10.1016/j.csda.2010.11.001. [p2]

R. E. Fay and R. A. Herriot. Estimates of income for small places: An application of James-Stein procedures to census data. *Journal of the American Statistical Association*, 74(366a):269–277, 1979. doi: 10.1080/01621459.1979.10482505. [p1, 3]

A. Gelman. Prior distributions for variance parameters in hierarchical models. *Bayesian Analysis*, 1(3): 515–534, 2006. doi: 10.1214/06-BA117A. [p5]

W. González-Manteiga, M. J. Lombardía, I. Molina, D. Morales, and L. Santamaría. Bootstrap mean squared error of a small-area EBLUP. *Journal of Statistical Computation and Simulation*, 78(5):443–462, 2008. doi: 10.1080/00949650601141811. [p2]

R. Janicki. Properties of the beta regression model for small area estimation of proportions and application to estimation of poverty rates. *Communications in Statistics - Theory and Methods*, 49(9): 2264–2284, 2020. doi: 10.1080/03610926.2019.1570266. [p2, 3, 4]

L. Knorr-Held. Bayesian modelling of inseparable space-time variation in disease risk. *Statistics in Medicine*, 19(17–18):2555–2567, 2000. doi: 10.1002/1097-0258(20000915/30)19:17/18<2555::AID-SIM587>3.0.CO;2-#. [p5]

A.-K. Kreutzmann, S. Pannier, N. Rojas-Perilla, T. Schmid, M. Templ, and N. Tzavidis. The R package **emdi** for estimating and mapping regionally disaggregated indicators. *Journal of Statistical Software*, 91(7):1–33, 2019. doi: 10.18637/jss.v091.i07. [p2]

B. G. Leroux, X. Lei, and N. Breslow. Estimation of disease rates in small areas: A new mixed model for spatial dependence. In *Statistical Models in Epidemiology, the Environment, and Clinical Trials*, pages 135–178. Springer-Verlag, New York, 2000. doi: 10.1007/978-1-4612-1284-3_4. [p5]

Z. R. Li, Y. Hsiao, J. Godwin, B. D. Martin, J. Wakefield, and S. J. Clark. Changes in the spatial distribution of the under-five mortality rate: Small-area analysis of 122 DHS surveys in 262 subregions of 35 countries in Africa. *PLOS ONE*, 14(1):e0210645, 2019. doi: 10.1371/journal.pone.0210645. [p2]

Y. Marhuenda, I. Molina, and D. Morales. Small area estimation with spatio-temporal Fay-Herriot models. *Computational Statistics & Data Analysis*, 58:308–325, 2013. doi: 10.1016/j.csda.2012.09.002. [p1, 4]

I. Molina and Y. Marhuenda. **sae**: An R package for small area estimation. *The R Journal*, 7(1):73–87, 2015. doi: 10.32614/RJ-2015-007. [p2, 3]

A. Petrucci and N. Salvati. Small area estimation for spatial correlation in watershed erosion assessment. *Journal of Agricultural, Biological, and Environmental Statistics*, 11(2):169–182, 2006. doi: 10.1198/108571106X110531. [p1, 3]

D. Pfeffermann. New important developments in small area estimation. *Statistical Science*, 28(1):40–68, 2013. doi: 10.1214/12-STS395. [p1]

N. G. N. Prasad and J. N. K. Rao. The estimation of the mean squared error of small-area estimators. *Journal of the American Statistical Association*, 85(409):163–171, 1990. doi: 10.1080/01621459.1990.10475320. [p3]

M. Pratesi, editor. *Analysis of Poverty Data by Small Area Estimation*. John Wiley & Sons, Chichester, United Kingdom, 2016. doi: 10.1002/9781118814963. [p1]

M. Pratesi and N. Salvati. Small area estimation: The EBLUP estimator based on spatially correlated random area effects. *Statistical Methods & Applications*, 17(1):113–141, 2008. doi: 10.1007/s10260-007-0061-9. [p1, 3]

J. N. K. Rao and I. Molina. *Small Area Estimation*. John Wiley & Sons, Hoboken, New Jersey, 2nd edition, 2015. doi: 10.1002/9781118735855. [p1, 2, 6]

A. Riebler, S. H. Sørbye, D. Simpson, and H. Rue. An intuitive Bayesian spatial model for disease mapping that accounts for scaling. *Statistical Methods in Medical Research*, 25(4):1145–1165, 2016. doi: 10.1177/0962280216660421. [p5]

H. Rue, S. Martino, and N. Chopin. Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 71(2):319–392, 2009. doi: 10.1111/j.1467-9868.2008.00700.x. [p2, 4]

A. Saei and R. Chambers. Out of sample estimation for small areas using area level data. Methodology Working Paper M05/11, Southampton Statistical Sciences Research Institute, University of Southampton, 2005. URL https://eprints.soton.ac.uk/14327/. [p4]

D. Simpson, H. Rue, A. Riebler, T. G. Martins, and S. H. Sørbye. Penalising model component complexity: A principled, practical approach to constructing priors. *Statistical Science*, 32(1):1–28, 2017. doi: 10.1214/16-STS576. [p2, 5]

S. H. Sørbye and H. Rue. Scaling intrinsic Gaussian Markov random field priors in spatial modelling. *Spatial Statistics*, 8:39–51, 2014. doi: 10.1016/j.spasta.2013.06.004. [p5]

M. Stefan and M. A. Hidiroglou. Benchmarked estimators for a small area mean under a onefold nested regression model. *International Statistical Review*, 89(1):108–131, 2021. doi: 10.1111/insr.12380. [p2, 6]

J. Wang, W. A. Fuller, and Y. Qu. Small area estimation under a restriction. *Survey Methodology*, 34 (1):29–36, 2008. URL https://www150.statcan.gc.ca/n1/pub/12-001-x/2008001/article/10619-eng.pdf. [p2]

Y. You and J. N. K. Rao. Small area estimation using unmatched sampling and linking models. *The Canadian Journal of Statistics*, 30(1):3–15, 2002. doi: 10.2307/3315862. [p2, 6]

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